

Machine Learning for Physicists - Homework 2 - Part II: Representing a structured function with a vision transformer

Consider a 2D grid of binary variables, $\{-1, 1\}$, arranged on a $s \times s$ lattice. The lattice is partitioned into non-overlapping 2×2 patches. Each patch contains 4 binary variables, so the total number of patches is

$$T = \frac{s^2}{4}.$$

For each patch $i = 1, \dots, T$, define the local configuration vector

$$\mathbf{x}_i = (\sigma_{i,1}, \sigma_{i,2}, \sigma_{i,3}, \sigma_{i,4})^T,$$

where $\sigma_{ij} \in \{-1, 1\}$ for $j = 1, 2, 3, 4$. We call the matrix of all σ_{ij} the matrix $\sigma \in \{\pm 1\}^{T \times 4}$. We also define the patch-indexing function

$$P(i) = \begin{cases} (i+1)\%T & \text{if } i\%(s/2) = 0, \\ (i+s/2)\%T + 1 & \text{otherwise.} \end{cases} \quad (1)$$

Consider the real-valued function

$$f : \{-1, 1\}^{T \times 4} \rightarrow \mathbb{R},$$

defined by

$$f(\sigma) = \sum_{i=1}^T \log \cos\left(\frac{\pi}{2} + \pi(\sigma_{i,4} + 3\sigma_{P(i),1})\right) + \log \cos\left(\frac{\pi}{2} + \pi(\sigma_{i,2} + 3\sigma_{i,3})\right). \quad (2)$$

The goal of this exercise is to show that the function in (2) can be represented *exactly* by a *single-layer* Vision Transformer (ViT)¹. However, we use a simplified and very specific architecture, where as a patch embedding we simply take local configuration vector $\mathbf{x}_i$. We use two attention heads and no layer norm or skip connections. As an MLP, we use one with the nonlinearity $\log \cos(\cdot)$. While its input weights are learned, the output weights are not.

In addition to the setting above, instead of the standard dot-product attention, we will use *factored attention*. In general, given token vectors $\mathbf{x}_i$ and $\mathbf{x}_j$, the context vector at position i is

$$\mathbf{A}_i = \sum_j \alpha_{ij} V \mathbf{x}_j. \quad (3)$$

In *factored* attention, the attention weights α_{ij} that make up $\mathbf{A}_i \in \mathbb{R}^{T \times T}$ are *learnable parameters* depending *only on position indices*, not on the input values. Therefore, rather than $\alpha_{ij} = \text{softmax}\left(\frac{\mathbf{x}_i^\top Q^\top K \mathbf{x}_j}{\sqrt{d}}\right)$ we have:

$$\alpha_{ij} = p_{ij}. \quad (4)$$

Factored attention removes queries and keys and directly parameterizes the attention map.

¹A schematic and general ViT architecture is shown in Fig. 1.

To summarize, our version of the vision transformer processes the input σ for $h = 2$ heads, $d = 4$ model dimension looks as follows:

1. Use $\mathbf{x}_i \in \mathbb{R}^d$ as the $i = 1 \dots T$ embedding vectors,
2. compute the value for each head $\mathbf{v}_i^h = V^h \mathbf{x}_i$ with $V^h \in \mathbb{R}^{d/2 \times d}$,
3. compute the attention matrix for each head $h = 1, 2$ as $\mathbf{A}_i^h = \sum_{j=1}^T \alpha_{ij}^h \mathbf{v}_j^h$ with $\alpha_{ij}^h \in \mathbb{R}$ and $i, j = 1 \dots T$,
4. concatenate the attention vectors for the two heads $\bar{\mathbf{x}}_i = \text{Concat}_h[\mathbf{A}_i^h] \in \mathbb{R}^d$,
5. apply the MLP including the non-linearity $\mathbf{z} = \sum_{i=1}^T \log \cos(W \bar{\mathbf{x}}_i + c) \in \mathbb{R}^d$ with $W \in \mathbb{R}^{d \times d}$ and $c \in \mathbb{R}$,
6. and the final result of the transformer is $f_{\text{transf}}(\sigma) = \sum_{\mu=1}^d z_{\mu}$.

Exercise (10 pts):

- a) How many parameters does the transformer have?
- b) Determine the values of V^h, α_{ij}^h, W , and c for which $f_{\text{transf}}(\sigma) = f(\sigma)$.

Hint: Assume that the value matrices for the two heads have the following shape:

$$V^{(1)} = \begin{pmatrix} 0 & 0 & 0 & V_{11}^{(1)} \\ 0 & V_{22}^{(1)} & V_{23}^{(1)} & 0 \end{pmatrix}, \quad V^{(2)} = \begin{pmatrix} V_{14}^{(2)} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (5)$$

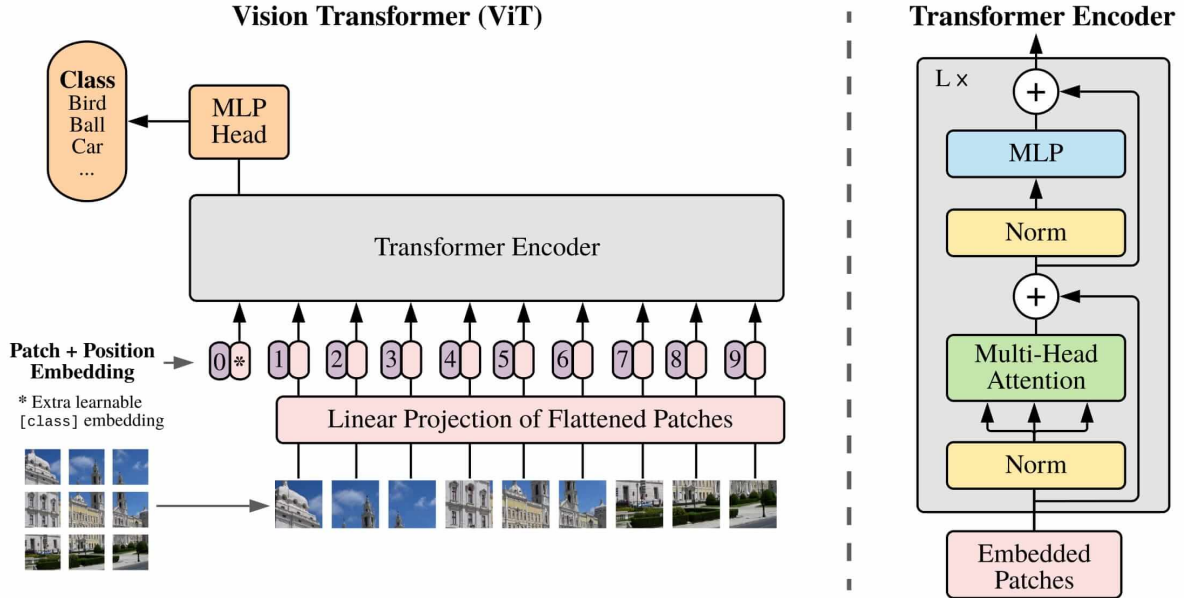


Figure 1: Schematic Vision Transformer architecture.